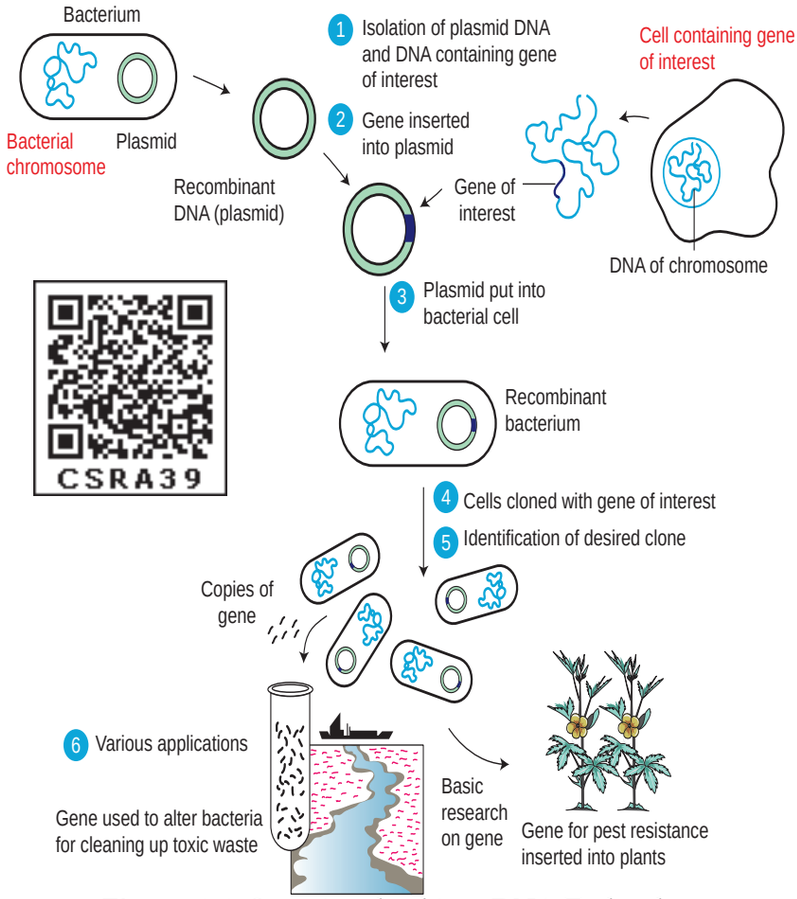


Figure 4.4: Steps involved in r-DNA Technology

into a carrier molecule called a **vector** that can self-replicate within the host cell.

- Selection of the transformed host cells is carrying the rDNA and allowing them to multiply thereby multiplying the rDNA molecule.
- The entire process thus generates either a large amount of rDNA or a large amount of protein expressed by the insert.
- Wherever vectors are not involved the desired gene is multiplied by PCR technique. The multiple copies are injected into the host cell protoplast or it is shot into the host cell protoplast by shot gun method.

4.5 Tools for Genetic Engineering

In order to generate recombinant DNA molecule, certain basic tools are necessary. The basic tools are enzymes, vectors and host organisms. The most important enzymes required for genetic engineering are the restriction enzymes, DNA ligase and alkaline phosphatase.

4.5.1 Restriction Enzymes

The two enzymes responsible for restricting the growth of bacteriophage in Escherichia coli were isolated in the year 1963. One was the enzyme which added methyl groups to DNA, while the other cut DNA. The latter was called restriction endonuclease. A **restriction** enzyme or **restriction endonuclease** is an enzyme that cleaves DNA into fragments at or near specific recognition sites within the molecule known as **restriction** sites. Based on their mode of action restriction enzymes are classified into Exonucleases and Endonucleases.

a. Exonucleases are enzymes which remove nucleotides one at a time from the end of a DNA molecule. e.g. Bal 31, Exonuclease III.

b. Endonucleases are enzymes which break the internal phosphodiester bonds within a

Restriction enzyme	Microbial source	Recognition sequence	Fragments
Alu I	Arthrobacter luteus	5'AG/CT3' 3'TC/GA5'	A-G C-T / T-C G-A — Blunt ends
BamHI	Bacillus amyloliquefaciens	5'G/GATCC3' 3'CCTAG/G5'	G / C-C-T-A-G — G-A-T-C-C / G — Sticky ends
EcoRI	Escherichia coli	5'G/AATTC3' 3'CCTAG/G5'	G / C-T-T-A-A — A-A-T-T-C / G — Sticky ends
HaeIII	Haemophilus aegyptus	5'GG/CC3' 3'CC/GG5'	G-G C-C / C-C G-G — Blunt ends
HindIII	Haemophilus influenza	5'A/AGCTT3' 3'TTCGA/A5'	A / T-T-C-G-A — A-G-C-T-T / A — Sticky ends

Table 4.1: Type II restriction enzyme with source, recognition and cleavage site.

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